DRAMCO

Miniature and customer friendly solution for tracking and locating insects

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Proposed Topic

Working Title

Miniature and customer friendly solution for tracking and locating insects

Background

The emergence of the Asian hornet has led beekeepers to invest numerous hours in locating nests. This is currently mainly done through labor-intensive triangulation methods, where hornets are lured and tracked from multiple locations. Existing miniature tracking solutions are either expensive, difficult to use, or limited in functionality for non-technical users. Typical systems rely on simple transmitters operating in the 142 MHz to 174 MHz band and require dedicated receivers with directional antennas.



This topic focuses on developing an affordable, miniature, and user-friendly system for tracking insects such as Asian hornets, with the aim of helping beekeepers locate nests faster and more efficiently.
The topic offers three clear research pathways, depending on the student's profile and interests:

- **BLE-based crowd-sourced tracking network (ICT-focused):** build on an existing transmitter and develop a crowd-sourced localization system using Bluetooth Low Energy, smartphones, gateways, or distributed receivers.
- **Improved transmitter design (Electronics-ICT-focused):** improve the current transmitter or tag design in terms of size, weight, range, battery life, cost, and usability.
- **Analog transmitter design (Electronics-focused):** investigate custom analog transmitter or transponder designs for extreme miniaturization, low power consumption, and simple low-cost tracking.

Current available technology:

- Two-stage transmitter[1]
- Single-stage transmitter[2]
- Custom Bluetooth protocol[3]
- Radar technology[4]

The goal of the research is to design a low-cost miniature transmitter or transponder together with a receiver that is easy to operate for non-technical people.

[1] https://journals.biologists.com/jeb/article/208/21/4063/15833/Miniaturization-0-2-g-and-evaluation-of-attachmentrada
[2] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8512766
[3] https://sensordrop.cs.washington.edu
[4] https://www.nature.com/articles/s41598-019-48511-8

Research Question

The main research questions are:

- Which technology can be used to design a transmitter or transponder with a weight below 160 mg?
- What is the most suitable battery technology?
- How will the signal propagate in urban, forest, and suburban environments?
- Can we design a user-friendly tag and receiver for a non-technical person?
- Can the receiver support direction finding with the chosen technology?
- Can a drone be used to follow the insect and help pinpoint the nest?
- How should direction and signal strength be visualized?

Final Output

The expected final output is a prototype of a low-cost insect tag or transponder with a consumer-friendly receiver.

Profile of the Master Student

Depending on the selected pathway, the topic is suited for a student with a profile in ICT, Electronics-ICT, or Electronics.

Main Activities

During this thesis, the following tasks are expected: literature review, technology comparison, system design, implementation, lab evaluation, and field evaluation.

Contact Person

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